

Warm-up 1

Find at least two solutions to the following system of linear equations over the real numbers.

$$\begin{aligned}w - 2x + y + z &= 0 \\ y + z &= 0\end{aligned}$$

Warm-up 2

Let's try out some games.

a) First, you get to pick any four distinct numbers from the set $\{1, 2, 3, \dots, 25\}$. Then I get to take whichever of your numbers I want and multiply them together. If the answer is a square number, I win. If not, you win.

For example, if you pick $\{2, 4, 6, 8\}$, I could win by making $2 \times 8 = 16$. Or just by picking 4.

Who do you think will win this game, if we both play strategically? Why?

b) Okay, now you have to pick any *ten* distinct numbers from the set $\{1, 2, 3, \dots, 25\}$. Then I get to take whichever of your numbers I want and multiply them together. Again, if the answer is a square number, I win. If not, you win.

Who do you think will win now? And why?

Advice from my former 20250 students:

“Although we only need to select four questions in each homework for correctness, we should still try to understand how to approach most of them.”

“Engage early, even if you have familiarity with linear algebra. Really don’t take too much advantage of the homework flexibility, even when it’s tempting to.”

“I made the mistake of passively reading the textbook, without writing notes to myself, the first few weeks of the quarter, and realized as we neared the midterm that I hadn’t quite digested the material as deeply as I should have. So, I would emphasize to future students to really take the reading seriously and to be sure not to brush over details while they engage in the reading. If while reading, one comes upon a confusion, note it down and ask for clarification as soon as you can. This way, one will be on top of the reading as well and have the understanding they need building steadily throughout the course. If one doesn’t do this, it can be easy to fall behind and have to take lots of hours later to catch up on reading and understanding. However again, try not to get too caught up on the details when assessments come around! For assessments, it’s important to simply do practice problems and understand key definitions.”

“I got much of my flex points through good hw and problem swap problems. This quarter was busy for me, and I admittedly didn’t spend as much time as I would have liked with flex opportunities, but they were a valuable experience.”

“The smaller flex options, like office hours and study groups, were nice incentives to keep up with the course. Also, I really enjoyed the large project option.”

“A lot of the ‘come up with an example’ type of problems I found difficult, but I think these did really help me understand the material.”

“Remember, there are two parts of the course you need to keep up with: theory AND computations. Quizzes are fast, but you have time for retakes. Do the hardest problems on the homework; it’ll help with exams. Have fun with flex projects and start early.”